

Two Types of Pressure Forces in the PPC Law of Gravity

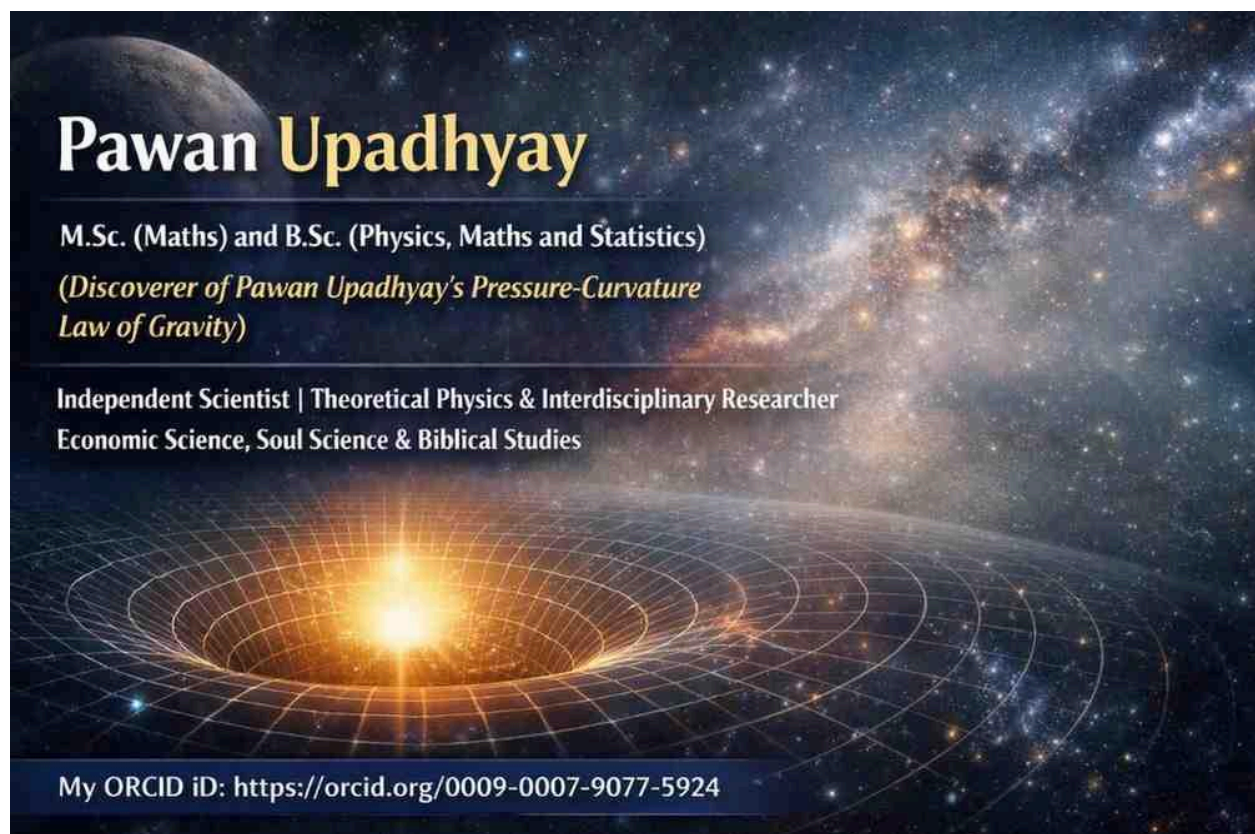
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Abstract

The Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity) introduces gravitational pressure as a fundamental physical quantity associated with the distribution of energy density in space. Within the PPC framework, gravitational pressure gives rise to two

distinct types of force manifestations: a volumetric force density acting throughout a region of space and a surface force acting on a boundary. This paper distinguishes these two quantities, establishes their mathematical definitions, and demonstrates their dimensional consistency.

1. Introduction

Pressure is a fundamental quantity in continuum physics and is capable of generating forces in different ways. A pressure field may produce a force density when its magnitude varies from one location to another, or it may produce a net force when acting directly upon a surface.

In the PPC Law of Gravity, gravitational pressure is represented by the gravitational pressure field P_g . The theory identifies two distinct pressure-force quantities:

1. **Gravitational Field Pressure Force Density**
2. **Gravitational Surface Pressure Force**

These quantities describe different physical manifestations of the same underlying gravitational pressure field.

2. Gravitational Pressure Field

The gravitational pressure field is related to the energy density by

$$[P_g = \omega E_d,]$$

where:

- P_g = gravitational pressure field,
- E_d = energy density,
- ω = equation-of-state parameter.

Since the equation-of-state parameter is dimensionless,

$$[\omega] = 1,]$$

the dimensions of gravitational pressure are identical to those of energy density:

$$[P_g]$$

[E_d]

$\frac{\text{J}}{\text{m}^3}$

$\frac{\text{N}}{\text{m}^2}$.]

Thus gravitational pressure possesses the SI unit Pascal (Pa).

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$$[\omega] = 1.$$

Energy density is defined as

$$E_d = \frac{E}{V},$$

where E is energy and V is volume.

Since

$$[E_d] = \frac{J}{m^3} = \frac{N}{m^2} = Pa,$$

the gravitational pressure field possesses the same dimensions and units as energy density:

$$[P_g] = Pa.$$

Therefore, the equation

$$P_g = \omega E_d$$

is dimensionally consistent.

3. Gravitational Field Pressure Force Density

A spatial variation in gravitational pressure generates a force density throughout space. In the PPC framework this quantity is defined by

$$[F = -\nabla P_g.]$$

where:

- F = gravitational field pressure force density.

The negative sign indicates that the force density is directed toward decreasing gravitational pressure.

Units and Dimensions

The gradient operator contributes one inverse length:

$$[\nabla = L^{-1}.]$$

Therefore,

$$[F]$$

$$[\nabla P_g]$$

$$\frac{\text{Pa}}{\text{m}}$$

$$\frac{\text{N}}{\text{m}^3}.]$$

The dimensional form is

$$[F]$$

$$M, L^{-2}, T^{-2}.]$$

Physical Interpretation

The gravitational field pressure force density represents a force distributed throughout a volume of space. It is a field quantity rather than a force acting on a specific surface.

3. Gravitational Field Pressure Force Density

A non-uniform gravitational pressure field produces a force density throughout space. In the PPC framework, this quantity is defined by

$$F = -\nabla P_g,$$

where:

- F = gravitational field pressure force density.

The negative sign indicates that the force density points toward decreasing gravitational pressure.

Units and Dimensions

The gradient operator contributes one inverse length:

$$[\nabla] = L^{-1}.$$

Therefore,

$$[F] = [\nabla P_g] = \frac{Pa}{m} = \frac{N}{m^3}.$$

The dimensional form is

$$[F] = M L^{-2} T^{-2}.$$

Physical Interpretation

The gravitational field pressure force density is a volumetric quantity. It represents the force distributed throughout a region of space due to spatial variations in gravitational pressure.

Substituting

$$P_g = \omega E_d$$

into the field equation gives

$$F = -\nabla(\omega E_d).$$

Applying the product rule gives

$$F = - (E_d \nabla \omega + \omega \nabla E_d) .$$

or equivalently

$$F = -E_d \nabla \omega - \omega \nabla E_d .$$

Physical Interpretation

This equation shows that the gravitational field pressure force density has **two contributions**:

1. Equation-of-state gradient contribution

$$F_{\omega} = -E_d \nabla \omega$$

arising from spatial variations of the equation-of-state parameter.

2. Energy-density gradient contribution

$$F_E = -\omega \nabla E_d$$

arising from spatial variations of energy density.

Therefore,

$$F = F_{\omega} + F_E.$$

Dimensional Check

Since ω is dimensionless,

$$[\nabla\omega] = L^{-1}.$$

Thus

$$[E_d \nabla\omega] = \left(\frac{N}{m^2}\right) \left(\frac{1}{m}\right) = \frac{N}{m^3}.$$

Similarly,

$$[\omega \nabla E_d] = (1) \left(\frac{N}{m^3}\right) = \frac{N}{m^3}.$$

Both terms have units

$$N \, m^{-3},$$

which are the correct units of field pressure force density.

The general PPC relation is

$$P_g = \omega E_d,$$

where

$$\omega = \omega(x, y, z, t)$$

may vary with position and time.

The gravitational field pressure force density is therefore

$$F = -\nabla(\omega E_d).$$

Applying the product rule:

$$F = -E_d \nabla \omega - \omega \nabla E_d.$$

This is the **general form**.

Special Case: Constant Numerical ω

If

$$\omega = \text{constant},$$

then

$$\nabla\omega = 0.$$

The first term vanishes, leaving

$$F = -\omega\nabla E_d.$$

Thus,

$$F = -\omega\nabla E_d$$

is a special case of the more general equation

$$F = -E_d\nabla\omega - \omega\nabla E_d.$$

For constant ω ,

$$F = -\omega \nabla E_d.$$

This relation shows that gradients in energy density generate gravitational field pressure force density within the PPC framework.

4. Gravitational Surface Pressure Force

When gravitational pressure acts directly on a surface of area A, a net force is produced.

This force is defined as

$$[F_p = P_g A.]$$

where:

- F_p = gravitational surface pressure force.

Units and Dimensions

Since

$$[[P_g]$$

$$\frac{N}{m^2},]$$

multiplication by area gives

$$[[F_p]$$

$$\left(\frac{N}{m^2}\right)m^2$$

$$N.]$$

The dimensional form becomes

$$[[F_p]$$

$$M,L,T^{-2}.]$$

Physical Interpretation

The gravitational surface pressure force represents the total force exerted by gravitational pressure on a surface or boundary.

4. Gravitational Surface Pressure Force

When gravitational pressure acts on a surface area A , it produces a net force on that surface.

This force is defined as

$$F_p = P_g A,$$

where:

- F_p = gravitational surface pressure force,
- A = surface area.

Units and Dimensions

Since

$$[P_g] = \frac{N}{m^2},$$

multiplication by area gives

$$[F_p] = \left(\frac{N}{m^2}\right) m^2 = N.$$

The dimensional form becomes

$$[F_p] = M L T^{-2}.$$

Physical Interpretation

The gravitational surface pressure force represents the total force exerted by gravitational pressure on a surface or boundary.

Unlike the field pressure force density, which is distributed throughout a volume, the surface pressure force acts directly on an area.

5. Relationship with Ordinary Force

In the PPC notation, ordinary force is identified with the gravitational surface pressure force:

$$[f = F_p.]$$

Consequently,

$$[f = P_g A.]$$

Both quantities possess identical dimensions and SI units:

[[f]

[F_p]

N.]

Thus the ordinary force experienced by a surface is interpreted as the surface manifestation of gravitational pressure.

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In PPC notation, ordinary force is identified with the gravitational surface pressure force:

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Therefore,

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Both quantities possess identical dimensions and units:

$$[f] = [F_p] = N.$$

Thus, the ordinary force experienced by a surface is interpreted as the surface manifestation of gravitational pressure.

6. Comparison of the Two Pressure Forces

Quantity	Meaning	Equation	SI Unit
F	Gravitational field pressure force density	$F = -\nabla P_g$	N/m^3
F_p	Gravitational surface pressure force	$F_p = P_g A$	N
f	Ordinary force	$f = F_p$	N

The two quantities are fundamentally different:

- F describes a force density distributed through volume.
- F_p describes a total force acting on a surface.
- f is identified with the surface pressure force.

Equation	SI Unit
$F = -\nabla P_g$	N/m^3
$F_p = P_g A$	N
$f = F_p$	N

7. Unified PPC Relations

The PPC framework may therefore be summarized by the relations

$$[P_g = \omega E_d,]$$

$$[F = -\nabla P_g,]$$

and

$$[f = F_p = P_g A.]$$

These equations connect energy density, gravitational pressure, field force density, and surface force within a single mathematical structure.

7. Unified PPC Relations

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$$P_g = \omega E_d,$$

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$$f = F_p.$$

These equations connect energy density, gravitational pressure, field force density, and surface force within a unified mathematical structure.

8. Conclusion

The PPC Law of Gravity distinguishes two physically different consequences of gravitational pressure.

The first is the **gravitational field pressure force density**

$$[F = -\nabla P_g,]$$

which acts throughout a volume of space and possesses units of N/m^3 .

The second is the **gravitational surface pressure force**

$$[F_p = P_g A,]$$

which acts on a surface and possesses units of N .

Since ordinary force is identified with the surface pressure force,

$$[f = F_p.]$$

The relations

$$[P_g = \omega E_d,]$$

$$[F = -\nabla P_g,]$$

and

$$[f = F_p = P_g A]$$

form a dimensionally consistent framework for describing both volumetric and surface manifestations of gravitational pressure in the PPC Law of Gravity.

8. Conclusion

The PPC Law of Gravity distinguishes two physically different manifestations of gravitational pressure.

The first is the **gravitational field pressure force density**

$$F = -\nabla P_g,$$

which acts throughout a volume of space and possesses units of

$$N\,m^{-3}.$$

The second is the **gravitational surface pressure force**

$$F_p = P_g A,$$

which acts on a surface and possesses units of

N .

Since ordinary force is identified with the surface pressure force,

$$f = F_p.$$

The gravitational pressure field itself is related to energy density through

$$P_g = \omega E_d,$$

where ω is a dimensionless equation-of-state parameter.

Together, the equations

$$P_g = \omega E_d,$$

$$F = -\nabla P_g,$$

$$F_p = P_g A,$$

and

$$f = F_p$$

provide a dimensionally consistent framework describing both volumetric and surface manifestations of gravitational pressure in the PPC Law of Gravity.